

30	B	$\lambda = \frac{\ln 2}{4.88 \times 10^{10}} = 1.4204 \times 10^{-11} \text{ year}^{-1}$ Using $N = N_0 e^{-\lambda t}$, $N_{\text{Rb}} = N_0 e^{-\lambda t}$ $N_{\text{Sr}} = N_0 - N_{\text{Rb}}$ $= N_0 - N_0 e^{-\lambda t}$ $= N_0 (1 - e^{-\lambda t})$ $\frac{N_{\text{Sr}}}{N_{\text{Rb}}} = \frac{N_0 (1 - e^{-\lambda t})}{N_0 e^{-\lambda t}}$ $= \frac{1 - e^{-\lambda t}}{e^{-\lambda t}}$ $= e^{\lambda t} - 1$

Qn	Ans	Solution
		$0.082 = e^{\lambda t} - 1$ $e^{\lambda t} = 1.082$ $t = \frac{\ln(1.082)}{1.4204 \times 10^{-11}}$ $= 5.55 \times 10^9 \text{ years}$